

HENDERSON, KY, USA

WMO: 724238

Lat: 37.800N

Lon: 87.683W

Elev: 387

StdP: 14.49

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53886

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.5	13.8	-1.9	5.0	10.7	2.7	6.4	14.8	25.1	45.1	23.2	42.7	6.3	320	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	93.3	76.4	91.2	75.9	90.0	75.4	79.5	90.7	78.2	88.9	76.8	86.9	7.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.4	139.7	86.8	74.8	132.4	85.4	73.1	124.9	83.1	43.2	91.1	41.8	88.8	40.4	86.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	18.8	16.8	-1.1	97.0	9.3	2.3	-7.9	98.7	-13.3	100.0	-18.6	101.3	-25.4	102.9		
(5)			-1.2	82.1	8.4	1.7	-7.2	83.4	-12.1	84.4	-16.8	85.4	-22.9	86.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	33.2	36.9	46.2	57.6	67.0	75.0	77.5	76.2	69.6	58.3	45.9	37.1		
	DBStd	18.03	11.83	11.31	10.73	9.21	7.85	5.45	4.68	4.92	7.12	8.98	9.46	10.16		
	HDD50	1782	535	386	204	34	1	0	0	0	0	23	185	414		
	HDD65	4419	986	787	590	253	74	4	0	1	36	249	574	865		
	CDD50	4271	14	20	85	263	529	751	851	812	588	280	63	15		
	CDD65	1432	0	0	6	32	137	305	387	348	174	41	2	0		
	CDH74	13779	0	1	42	290	1203	2909	3796	3349	1754	424	11	0		
	CDH80	5539	0	0	3	48	376	1183	1646	1444	721	117	1	0		
(14) Wind	WSAvg	7.0	8.7	8.6	8.7	8.9	7.3	5.7	4.5	4.2	4.8	6.6	7.7	8.0		
(15) Precipitation	PrecAvg	43.0	3.5	3.1	4.4	4.4	5.2	3.6	3.8	2.8	3.2	3.4	3.7	3.8		
	PrecMax	61.9	6.4	6.3	10.9	10.9	11.7	6.1	7.4	9.1	10.9	9.2	6.7	7.5		
	PrecMin	30.4	1.3	0.9	2.0	1.7	1.2	1.5	1.7	0.7	0.0	0.6	0.5	1.2		
	PrecStd	7.5	1.4	1.7	2.4	2.3	2.6	1.4	1.7	2.0	2.6	2.1	1.6	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.5	70.3	79.4	84.1	90.8	95.0	98.5	98.6	94.9	88.3	75.0	66.2		
		MCWB	58.8	60.9	64.1	67.1	72.7	76.8	79.0	78.3	74.0	71.7	61.9	60.0		
	2%	DB	62.4	64.2	72.9	81.3	87.9	91.4	92.9	93.1	91.0	82.5	70.1	61.2		
		MCWB	56.8	56.6	60.2	65.1	71.6	75.2	78.3	77.1	73.0	67.2	60.3	55.2		
	5%	DB	56.7	60.7	69.5	77.5	84.1	90.1	90.7	90.5	88.3	79.1	65.6	56.6		
		MCWB	51.9	53.7	59.0	63.2	69.5	74.7	77.7	76.1	72.1	64.4	57.6	51.5		
	10%	DB	51.5	54.7	63.7	73.0	81.5	87.7	88.5	88.1	83.9	74.0	61.9	52.4		
		MCWB	46.8	48.5	54.9	61.1	68.0	73.8	76.2	75.1	70.0	62.8	54.0	47.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	62.7	65.8	70.0	75.3	79.4	82.2	81.1	77.0	73.6	65.9	61.5		
		MCDB	63.9	68.5	76.9	81.2	86.0	91.2	92.6	91.9	90.0	85.2	74.0	65.7		
	2%	WB	57.5	58.7	62.2	67.5	73.1	77.7	80.3	79.0	75.2	70.6	61.8	56.6		
		MCDB	62.4	63.4	71.4	77.7	84.5	89.0	90.7	90.6	87.6	79.1	67.6	60.7		
	5%	WB	52.1	54.3	59.2	65.2	71.3	76.2	78.7	77.6	73.3	67.2	58.3	52.1		
		MCDB	56.3	59.8	67.2	74.7	81.9	87.1	89.0	88.4	84.7	75.2	64.4	56.3		
	10%	WB	46.0	48.6	55.9	62.7	69.3	74.6	77.4	76.0	71.4	64.2	54.7	47.7		
		MCDB	49.6	54.8	63.2	71.5	78.9	84.9	86.9	85.5	81.0	72.8	61.5	51.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	16.8	18.9	21.3	19.7	19.7	19.8	21.3	23.8	22.9	18.9	14.8		
		MCDBR	19.9	23.9	24.9	25.2	23.3	22.9	22.4	24.5	27.9	27.1	24.7	19.0		
	5% WB	MCWBR	17.4	18.7	15.9	14.6	11.4	10.7	11.6	12.0	12.8	14.4	17.1	16.0		
		MCWBR	18.3	21.8	21.6	21.3	20.4	19.7	20.0	21.7	23.7	22.5	20.4	17.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.328	0.368	0.406	0.438	0.451	0.478	0.484	0.415	0.363	0.330	0.311		
		taud	2.457	2.402	2.351	2.250	2.212	2.198	2.135	2.106	2.292	2.430	2.469	2.475		
	Ebn at Noon	Ebn at Noon	275	286	283	277	269	265	256	251	264	270	267	269		
		Edh at Noon	26	31	36	43	45	46	49	49	38	30	26	24		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	635	856	1191	1578	1773	2020	1938	1783	1482	1104	753	547		
	RadStd	RadStd	66	101	109	91	154	125	124	93	104	116	81	50		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+139

Nomenclature: See separate page